

Biogeochemical signatures preserved in ancient siliceous sediments; new perspectives to Triassic radiolarian bedded chert compositions

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Major and minor elements of successively collected 70 Middle Triassic radiolarian bedded cherts from the Mino Belt in central Japan were studied. In order to reveal pre-diagenetic, primary geochemical signatures related to marine biogeochemical cycles, composite data of chert-shale couplets in addition to raw data were examined. The results show that Mn, Cu, Sr, Ba and P were supplied significantly by non-lithogenic excess fractions. Positive correlation with SiO₂/TiO₂ suggests that accumulations of Ba, Sr, Mn and Cu were closely related to sedimentation of biogenic silica; Mn and Cu were probably incorporated into oxides, whereas Ba into barite. Relationship between SiO₂/TiO₂ and excess-Ba, proxy for surface productivity implies that regeneration of silica from sinking radiolarian tests was not significant. This may have been caused by an expected relatively short travel time of radiolarian tests from the ocean surface to the bottom, reflecting their large size compared with diatoms. Positive correlations between CaO and P₂O₅ imply that apatite species are host phases for P and Ca. Although conodont composed of apatite is abundant in both chert and shale, their CaO/P₂O₅ ratio differs markedly from those of cherts and shales. Other apatite species, possibly carbonate apatite of diagenetic origin, are present in the bedded cherts and contribute to excess-P accumulation. Early diagenesis is believed to have modified only slightly primary geochemical features of the studied bedded cherts. This suggests relatively oxic depositional environment for the bedded cherts, under which regeneration of redox sensitive elements were not active.

Keywords: Triassic, radiolarian bedded chert, trace metal, accumulation, diagenesis

INTRODUCTION

Radiolarian bedded cherts are characterized by rhythmic alternation of silica-rich chert beds several cm thick and shale layer mm thick. The fine-grained siliceous sedimentary rocks are originated from marine biogenic sediments (radiolarian and diatomaceous ooze) and are common in the circum-Pacific orogenic belts (e.g., Hein and Parrish, 1987). They are composed originally of opaline biogenic silica, clastic lithogenic materials and authigenic components from seawater and occasionally hydrothermal solution (Pollock, 1987; Shimizu *et al.*, 2001; Yamamoto, 1983). Their geochemistry has been extensively studied in the context of the origin, the depositional environment (pelagic or hemipelagic), the geologic evolution of orogenic belts and the global environmental change (e.g., Kato *et al.*, 2002; Kunimaru *et al.*, 1998; Matsumoto and Iijima, 1983; Murray *et al.*, 1991, 1992b; Shimizu and Masuda, 1977; Sugisaki *et al.*,

1982). On the other hand, bedded chert geochemistry has been examined rarely from the point of view of marine geochemical cycle involving biological uptake of elements in the ocean surface, and scavenging by settling biogenic particles and by terrigenous particulate matter (e.g., Bishop, 1988; Boyle, 1981; Bruland *et al.*, 1991; Kuss and Kremling, 1999; Masuzawa *et al.*, 1989; Murray and Leinen, 1993; Nameroff *et al.*, 2002; Piper, 1994; Turekian, 1977).

In this study, Triassic bedded chert in the Mino Belt, central Japan was studied in the aim of revealing signatures related to marine biogeochemical cycles. The bedded cherts have been metamorphosed to the very low grade, and thus redistribution of elements by metamorphism is basically negligible. However, diagenetic effects that might have modified their primary mineral and geochemical characteristics should be taken into account (Murray *et al.*, 1992a, b; Murray, 1994; Tada, 1991; von Rad and Röscher, 1974). In order to reveal primary geochemical signatures through suspected diagenetic overprints, we performed complete successive collection of 70 samples and calculated composite data of chert-shale couplet. We attempt to interpret geochemical characteristics of bedded cherts in analogy with modern marine

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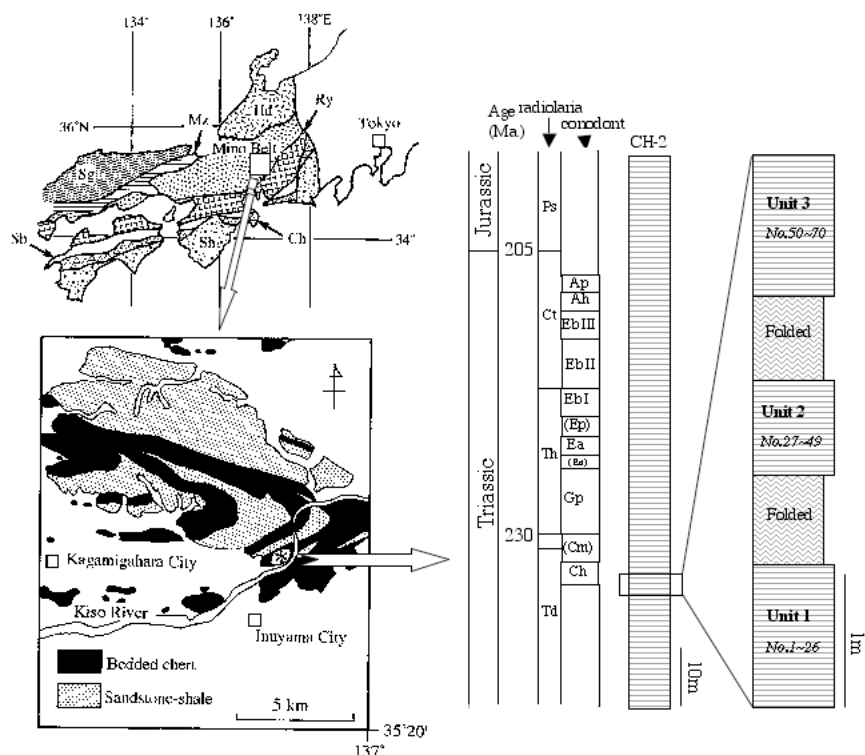


Fig. 1. Geological map of central-southwestern Japan (modified from Sugitani and Mimura, 1998) and stratigraphic column of the studied bedded chert sequence with radiolarian and conodont biostratigraphy (Matsuda and Isozaki, 1991). Hd; Hida Beld, Sg; Sangun Belt, Mz; Maizuru Belt, Ry; Ryoke Belt, Ch; Chichibu Belt, Sb; Sambagawa Belt, Sh; Shimanto Belt.

geochemical cycles that are controlled significantly by biogenic processes. In particular, accumulation process of non-lithogenic “excess” (biogenic and hydrogenous) components is addressed, providing new perspectives to geochemistry of ancient biogenic siliceous sediments.

GEOLOGICAL AND GEOCHEMICAL BACKGROUND AND SAMPLING LOCATION

The Mino Belt is one of the major geological units in Japan (Fig. 1). It consists of chaotically mixed Permian to Triassic radiolarian bedded cherts, limestones, basalts, and Jurassic turbidite sandstones and shales (Wakita, 1988). Depositional environment of the radiolarian bedded cherts in this belt has long been debated (Matsuda and Isozaki, 1991; Shimizu *et al.*, 2001; Sugisaki *et al.*, 1982; Sugitani and Mimura, 1998; Yamamoto, 1983). Matsuda and Isozaki (1991) claimed that bedded cherts had deposited on the pelagic ocean floor far from continents, based on the lack of coarse terrigenous detritus and assumed very low sedimentation rate of the bedded cherts. On the other hand, some geochemists (e.g., Shimizu *et al.*, 2001; Sugisaki *et al.*, 1982; Sugitani and Mimura, 1998; Yamamoto, 1983) have suggested the deposition in hemi-pelagic regions, based on MnO/TiO₂ values, and Rb-

Sr and Sm-Nd isotopic data and REE abundances.

Around the sampling site in Kagamigahara City, Gifu Prefecture, four units of Triassic-Jurassic bedded cherts (CH1~4) (Yao *et al.*, 1980) are well exposed along the Kiso River (Fig. 1). Samples for the present study were collected from the unit CH2, where the color of bedded chert changes drastically from gray~black to brick-red in an ascending stratigraphic order. The gray~black cherts in the lower horizon are generally enriched in carbonaceous matter and sulfide (mainly pyrite), whereas the upper brick-red cherts lack carbonaceous matter and contain disseminated hematite particles (Kubo *et al.*, 1996). This clear stratigraphic color and compositional change in the bedded chert sequence reflects the change of depositional environment (Isozaki, 1997; Sugitani and Mimura, 1998). The gray~black carbonaceous cherts had deposited under the anoxic environment, whereas the brick-red cherts studied here are assumed to have deposited under the suboxic to oxic environment. There has been no evidence reported for influence of hydrothermal activities in the studied bedded chert. We collected 45 cherts and 25 shales from well bedded three units in this brick-red chert section. Complete “bed to bed” sampling was performed, with data of representative thickness of each bed.

Table 1. Averages and standard deviations for raw data

Raw-data	Unit-1		Unit-2		Unit-3	
	Chert (<i>n</i> = 16)	Shale (<i>n</i> = 10)	Chert (<i>n</i> = 16)	Shale (<i>n</i> = 7)	Chert (<i>n</i> = 13)	Shale (<i>n</i> = 8)
SiO ₂ (wt%)	92.31 ± 2.71	67.11 ± 2.19	91.23 ± 2.73	65.67 ± 3.03	92.40 ± 2.37	74.63 ± 7.27
TiO ₂	0.125 ± 0.050	0.61 ± 0.05	0.138 ± 0.051	0.66 ± 0.06	0.117 ± 0.044	0.50 ± 0.15
Al ₂ O ₃	2.70 ± 1.00	12.89 ± 1.16	2.80 ± 1.10	13.30 ± 1.24	2.35 ± 0.82	10.21 ± 2.86
Fe ₂ O ₃ *	1.32 ± 0.51	5.97 ± 0.51	1.46 ± 0.53	6.58 ± 0.52	1.26 ± 0.46	5.01 ± 1.45
MnO	0.052 ± 0.030	0.11 ± 0.01	0.063 ± 0.022	0.15 ± 0.02	0.053 ± 0.023	0.12 ± 0.03
MgO	0.49 ± 0.16	2.26 ± 0.17	0.54 ± 0.20	2.45 ± 0.23	0.44 ± 0.18	1.92 ± 0.53
CaO	0.12 ± 0.01	0.80 ± 0.22	0.13 ± 0.06	0.43 ± 0.13	0.10 ± 0.02	0.47 ± 0.29
Na ₂ O	0.15 ± 0.10	0.45 ± 0.15	0.11 ± 0.10	0.37 ± 0.11	0.02 ± 0.04	0.13 ± 0.08
K ₂ O	0.88 ± 0.37	4.63 ± 0.35	0.90 ± 0.42	4.88 ± 0.48	0.74 ± 0.28	3.81 ± 1.09
P ₂ O ₅	0.034 ± 0.006	0.46 ± 0.14	0.044 ± 0.040	0.24 ± 0.07	0.028 ± 0.011	0.28 ± 0.18
H ₂ O*	0.99 ± 0.31	3.06 ± 0.24	1.08 ± 0.28	3.51 ± 0.34	1.02 ± 0.29	2.49 ± 0.77
C _{total}	0.01 ± 0.01	0.01 ± 0.01	0.00 ± 0.01	0.03 ± 0.05	0.01 ± 0.02	0.01 ± 0.01
S _{total}	0.00 ± 0.01	0.00 ± 0.01	0.00 ± 0.00	0.00 ± 0.01	0.00 ± 0.00	0.00 ± 0.00
Cr (ppm)	14.6 ± 7.2	59.8 ± 6.2	18.7 ± 8.5	62.5 ± 5.13	21.8 ± 9.0	46.7 ± 13.1
Co	3.5 ± 1.6	15.7 ± 2.2	4.4 ± 1.8	14.7 ± 2.9	5.2 ± 1.9	13.1 ± 2.9
Cu	50.3 ± 19.3	30.9 ± 5.2	53.6 ± 22.8	24.4 ± 3.50	43.2 ± 18.2	24.5 ± 5.1
Zn	30.1 ± 10.7	138.9 ± 21.6	35.6 ± 13.4	157.3 ± 20.2	31.1 ± 14.2	121.8 ± 33.2
Pb	6.6 ± 5.0	21.0 ± 4.9	6.6 ± 6.6	19.9 ± 4.7	7.6 ± 5.9	16.6 ± 5.1
V	17.8 ± 4.5	66.0 ± 12.8	19.0 ± 4.7	49.0 ± 8.1	15.1 ± 5.7	38.4 ± 11.3
Sr	38 ± 6	54 ± 4	39 ± 6	51 ± 3	33 ± 4	38 ± 6
Ba	121 ± 35	407 ± 33	129 ± 40	425 ± 37	115 ± 36	304 ± 78
Zr	27 ± 11	134 ± 12	30 ± 12	145 ± 11	26 ± 9	110 ± 31

*Total Fe and H as Fe₂O₃ and H₂O, respectively.

METHOD

Chemical analyses

Major elements were analyzed using an X-ray fluorescence spectrometer (XRF) by the method of Sugisaki *et al.* (1977). The fusion glass for XRF analysis was made from the mixture of powdered sample and flux (Li₂B₄O₇) in the proportion of 1:5.

For analyses of trace elements (Cr, Co, Cu, Zn, Pb and V), about 100 mg powdered samples were decomposed by 0.5 ml-HF, 0.5 ml-HNO₃ and 1.0 ml-HClO₄ in a Teflon bomb at 160°C for 10 hours according to Sugitani *et al.* (2002). Sample solution was evaporated to dryness, and residues were dissolved in 100 ml 1% HNO₃ solution. Trace elements in sample solution were analyzed by a graphite-furnace atomic absorption spectrometer. Accuracy and precision for the analyzed elements are better than 5%. Analyses of Sr, Ba and Zr were operated using XRF, with accuracy and precision of better than 10% (Sugitani and Mimura, 1998). Pressed discs were made from the mixture of powdered sample and binder in the proportion of 1:1. Total C, H, and S were analyzed using an elemental analyzer. Average and standard deviation of

raw data for each unit are shown in Table 1*.

Additionally, we performed qualitative and semi-quantitative microanalyses of some authigenic phases such as sulphate and phosphate using an electron microprobe analyzer (JEOL JXA-8800R at Nagoya University). Accelerating voltage and specimen current for the analyses were 15 kv and 12 nA, respectively. The beam diameter was 5 µm. The ZAF method was employed for matrix correction. The results will be presented in the later sections.

Data handling

Diagenetic overprints that potentially disturb primary geochemical signatures of marine sediments involve remobilization of redox-sensitive elements such as Mn during early diagenesis (Bonatti *et al.*, 1971; Froelich *et al.*, 1979; Klinkhammer *et al.*, 1982; Sawlan and Murray, 1983) and later diagenetic elemental redistribution. The later diagenetic event is represented by dissolution and re-precipitation of silica (radiolarian tests), which could

*Whole data set is available upon request to the corresponding author (K. Sugitani).

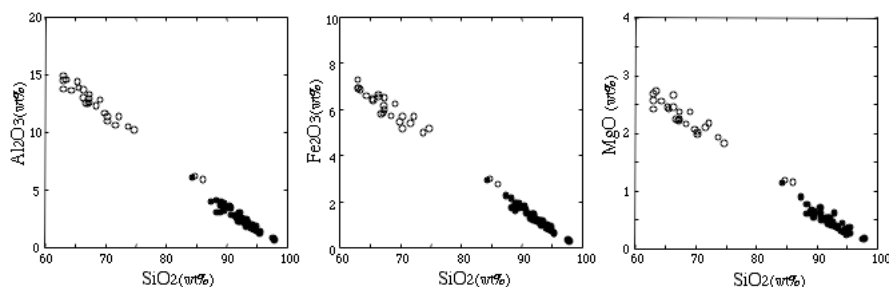


Fig. 2. SiO_2 - element plots for raw data of major elements. Open circle shows shale and closed one does chert.

be accompanied with significant redistribution of other elements (Murray *et al.*, 1992a, b). At least the effect of later diagenetic elemental redistribution could be minimized by data handling (re-calculating composite data of chert-shale couplet), as shown in detail below.

Composite data of each chert-shale couplet is given by calculating weighted average of successive chert-shale couplets in the following equation;

$$E_{\text{comp}} = (E_{\text{ch}} * D_{\text{ch}} * T_{\text{ch}} + E_{\text{sh}} * D_{\text{sh}} * T_{\text{sh}}) / (D_{\text{ch}} * T_{\text{ch}} + D_{\text{sh}} * T_{\text{sh}})$$

where E_{comp} is composite elemental concentration (wt%) of given chert-shale couplet; E_{ch} and E_{sh} show elemental concentrations (wt%) in chert and adjacent shale-partings, respectively; D and T show density and thickness in the same manner. Because direction of diagenetic migration of elements is equivocal, we calculated two composite data from one chert-shale-chert sequence, like calculation of “moving average”. Obtained composite data virtually represent composition of bulk sediment deposited during one cycle of deposition of siliceous and clay-rich sediments, assuming that the later diagenetic elemental redistribution occurred in a closed system.

RESULTS AND DISCUSSIONS

Comparison of raw data of cherts and shales, and its limits

Most representative geochemical feature of bedded cherts studied here is contrastive chemical compositions of cherts and their shale-partings (Table 1). Cherts are enriched in SiO_2 (~90%) and depleted in many of other elements, compared with shale-partings. This feature is more or less attributed to “dilution” with nearly pure silica of radiolarian tests, as exemplified by SiO_2 - Al_2O_3 , SiO_2 - Fe_2O_3 and SiO_2 - MgO plots (Fig. 2).

In order to compensate this dilution effect, normalization of constituent elements by Al or Ti have often been used (e.g., Murray *et al.*, 1992a; Murray, 1994; Sugisaki *et al.*, 1982; Yamamoto, 1983), based on that Al and Ti are incorporated mostly in lithogenic fraction and are not mobile during diagenesis. By using element/ TiO_2 or ele-

Table 2. TiO_2 -normalized data of bedded cherts

	Chert ($n = 45$)	Shale ($n = 25$)
SiO_2	901.5 ± 629.0	127.1 ± 54.8
Al_2O_3	20.97 ± 1.63	20.7 ± 0.64
Fe_2O_3	10.66 ± 0.28	9.94 ± 0.23
MnO	0.51 ± 0.34	0.21 ± 0.03
MgO	3.99 ± 0.73	3.77 ± 0.17
CaO	1.05 ± 0.67	1.00 ± 0.45
Na_2O	0.82 ± 1.09	0.54 ± 0.27
K_2O	6.56 ± 0.57	7.57 ± 0.26
P_2O_5	0.31 ± 0.22	0.57 ± 0.27
Cr	145.3 ± 64.2	95.9 ± 4.3
Co	34.7 ± 11.8	25.2 ± 4.3
Cu	531.9 ± 507.4	49.3 ± 21.5
Zn	259.2 ± 46.7	236.7 ± 23.1
Pb	51.9 ± 33.3	32.9 ± 6.4
V	148.5 ± 45.8	90.5 ± 25.3
Sr	332.1 ± 144.8	83.2 ± 11.2
Ba	1011.6 ± 206.7	647.1 ± 40.0
Zr	217.7 ± 18.1	221.1 ± 4.5

ment/ Al_2O_3 ratios, therefore, we could compare degree of elemental enrichment between bedded cherts and non-siliceous marine sediments.

TiO_2 -normalized values of cherts and shales are shown in Table 2, which indicates that the bedded chert geochemistry cannot always be explained solely by mixing of lithogenic detrital component and pure silica. TiO_2 -normalized values of some elements such as Cu and Sr in cherts are higher than those in shales. These elements are substantially enriched in cherts. Additionally they, if not all, show positive correlation with $\text{SiO}_2/\text{TiO}_2$ (Fig. 3). It should be emphasized that the positive correlations in Fig. 3 are not results of different degree of chemical weathering between chert and shale, because the correlations are defined mainly by scattering of chert data. One may, therefore, take this relationship as evidence for that the accumulation of these elements was associated with deposition of radiolarian tests. However, this may not always be the case, since the positive relationship with $\text{SiO}_2/$

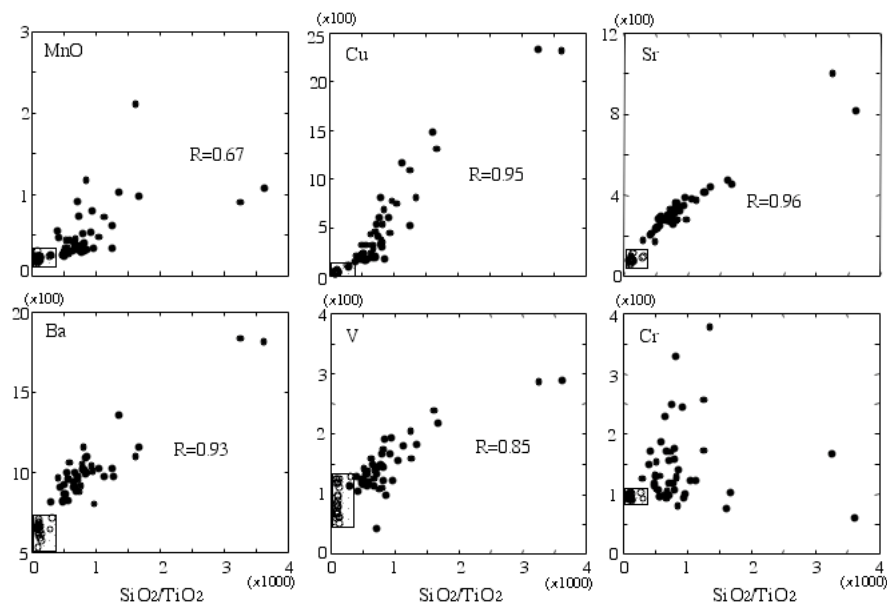


Fig. 3. $\text{SiO}_2/\text{TiO}_2$ - element/ TiO_2 plots for raw data of selected elements showing enrichment in cherts relative to shales on TiO_2 -normalization basis. Open circle shows shale and closed one does chert. The square shows the area for shales.

TiO_2 can alternatively be explained by later diagenetic fractionation between cherts and shales. The diagenesis involves dissolution and re-precipitation of radiolarian tests, which are *at least partially* responsible for the present contrastive silica contents between cherts and shales (De Wever and Baudin, 1996; Tada, 1991; Murray *et al.*, 1992b). Mobilization of elements from proto-shale bed to proto-chert bed may have resulted in the observed features of the raw data, that is, higher element/ TiO_2 ratios in cherts than in shales, and the positive correlation with $\text{SiO}_2/\text{TiO}_2$.

The two important features of the bedded chert discussed above are summarized as 1) some elements are enriched in cherts than in shales on TiO_2 -normalization basis and 2) positive correlation can be seen between TiO_2 -normalized values of the enriched elements and $\text{SiO}_2/\text{TiO}_2$. In order to clarify the validity of these features, the following approaches using composite data are taken. First, composite data of chert-shale couplets are compared with assumed lithogenic components in bedded cherts. This comparison would provide information about accumulation of elements as non-lithogenic components, that is excess fraction. Second, the relationships between element/ TiO_2 and $\text{SiO}_2/\text{TiO}_2$ for the composite data are examined. This would provide constraints on validity of the positive correlations with $\text{SiO}_2/\text{TiO}_2$ observed for some elements in the raw data.

Analyses of composite data of chert-shale couplets

Average compositions of composite data of chert-shale

couplets (see Section “Method”) and three hypothetical compositions of lithogenic components in the bedded chert are shown in Table 3. Listed hypothetical compositions of lithogenic detrital component are average shale composition (Wedepohl, 1995), upper crustal composition (Krauskopf and Bird, 1995) and more specific model composition based on Shimizu *et al.* (2001). Shimizu *et al.* (2001) proposed that detrital components in Triassic bedded cherts in the Mino Belt are composed dominantly of typical continental materials represented by aeolian loess (90%) with minor amount of granitic materials (10%). Enrichment factors relative to these three hypothetical lithogenic components are also shown in Table 3. Enrichment factor (*E.F.*), which is defined by $((\text{element}/\text{Ti})_{\text{chert-shale}}/(\text{element}/\text{Ti})_{\text{lithogenic/igneous}})$, is modified from equation of Calvert and Pedersen (1993), allowing for the possibility of biogenic uptake of Al (Banakar *et al.*, 1998; Murray and Leinen, 1996).

Non-lithogenic excess components in the bedded cherts
As shown in Table 3, the enrichment factors calculated to the three different estimates of lithogenic components are not always the same. This is due to inevitable uncertainties in hypothetical compositions of lithogenic components. Nevertheless, it is apparent that accumulation of Mn, K, P, Cu and Zn could not be explained solely by contribution of lithogenic detrital components. These elements are supposed to have been accumulated in a significant portion as non-lithogenic components. As a rough estimation, 50% or more of these enriched elements were supplied as non-lithogenic excess fractions.

Table 3. Averages and ranges of integrated data of chert-shale couplets, and compositions of three hypothetical lithogenic components

	Integrated data	Lithogenic components and enrichment factors (<i>E.F.</i>)					
	Average and range	LGM	<i>E.F.</i>	Shale	<i>E.F.</i>	Crust	<i>E.F.</i>
SiO ₂ (wt%)	85.07 (75.50~88.72)	66.00	3.37	63.93	4.30	66.26	2.62
TiO ₂	0.26 (0.19~0.46)	0.68	1.00	0.84	1.00	0.53	1.00
Al ₂ O ₃	5.48 (3.80~9.23)	13.53	1.06	16.54	1.07	14.93	0.75
Fe ₂ O ₃	2.69 (1.96~4.72)	4.91	1.43	7.38	1.18	4.51	1.22
MnO	0.08 (0.05~0.13)	0.09	2.32	0.12	2.15	0.07	2.33
MgO	1.01 (0.73~1.66)	2.15	1.23	2.72	1.20	2.29	0.90
CaO	0.25 (0.10~0.46)	7.87	0.08	3.38	0.24	4.2	0.12
Na ₂ O	0.18 (0.03~0.44)	1.45	0.32	1.41	0.41	3.53	0.10
K ₂ O	1.91 (1.22~3.25)	2.83	1.77	3.51	1.76	3.52	1.11
P ₂ O ₅	0.12 (0.05~0.25)	0.16	1.96	0.14	2.77	0.16	1.53
Cr (ppm)	29 (17~47)	90	0.84	90	1.04	36	1.64
Co	7 (4~12)	16	1.14	19	1.19	12	1.21
Cu	45 (25~79)	32	3.68	45	3.23	14	6.55
Zn	64 (43~119)	75	2.23	95	2.18	53	2.46
Pb	11 (6~24)	26	1.11	20	1.78	17	1.32
V	29 (14~44)	97	0.78	130	0.72	54	1.09
Sr	41 (30~52)	255	0.42	300	0.44	323	0.26
Ba	201 (149~329)	504	1.04	580	1.12	682	0.60
Zr	58 (42~103)	313	0.48	160	1.17	242	0.49

Note: LGM shows lithogenic compositions calculated based on loess-granite model (Shimizu et al., 2001). Data for this calculation were quoted from Gallet et al. (1996) and Condé (1993) for major elements, and Liu et al. (1993), Taylor et al. (1983) and Krauskopf and Bird (1995) for minor elements. Max values in available data of minor elements were selected for loess components. Data for shale and crust is from Krauskopf and Bird (1995) and Wedepohl (1995), respectively.

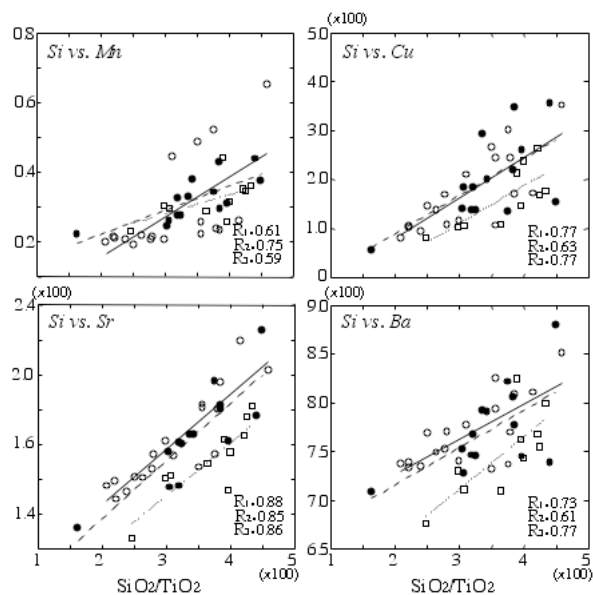


Fig. 4. SiO₂/TiO₂ - element/TiO₂ plots for composite data. Open circles, closed circles and open squares show data from Unit 1, Unit 2 and Unit 3, respectively; R1, R2 and R3 show correlation coefficients for individual units. The composite data have SiO₂/TiO₂ values from 162 to 459, except 6 data having SiO₂/TiO₂ values higher than 500. These 6 data apart from an overall trend are eliminated from these plots.

Inter-element relationships of integrated data Table 4 and Fig. 4 show that in composite data set, weak to moderate positive correlations can be seen for TiO₂-normalized values of some elemental pairs (Si-Mn, Si-Cu, Si-Ba, Si-Sr, Mn-Cu, Mn-Mg, Sr-Ba, P-Ca). Since composite-data calculation would artificially reduce original variations in element/TiO₂, the relationships of MnO/TiO₂, Cu/TiO₂, Ba/TiO₂, Sr/TiO₂ with SiO₂/TiO₂ shown in Fig. 4 should not be underestimated, although correlation coefficients are not always high. As discussed in the later section, the relationships seen in composite data set, combined with those in raw data set, would be diagnostic for the close association of element accumulation with biogenic silica (radiolarian tests).

Authigenic and biogenic phases in bedded cherts

By microprobe analyses in addition to usual petrographic examinations, we identified some authigenic and biogenic phases in bedded cherts. They are apatite (conodont) and possibly barite, which will be further described below.

Apatite (conodont) Conodont is the most representative phase for phosphate identified under the microscope. They are contained in both cherts and shales. Analytical results of conodont are shown in Table 5. There can be seen no systematic difference in compositions between conodonts

Table 4. Correlation coefficients for integrated data (TiO₂-normalized)

	SiO ₂	Fe ₂ O ₃	MnO	MgO	CaO	P ₂ O ₅	Co	Cu	Zn	V	Sr	Ba
SiO ₂	1.00											
Fe ₂ O ₃	0.25	1.00										
MnO	0.62	0.32	1.00									
MgO	0.39	0.26	0.82	1.00								
CaO	0.07	-0.28	0.11	0.26	1.00							
P ₂ O ₅	-0.02	-0.25	0.07	0.24	0.98	1.00						
Co	0.12	0.59	0.10	0.17	-0.04	0.01	1.00					
Cu	0.66	-0.02	0.67	0.46	0.05	-0.02	-0.22	1.00				
Zn	0.03	0.49	0.35	0.41	-0.16	-0.10	0.37	0.00	1.00			
V	0.18	-0.49	0.32	0.28	0.36	0.35	-0.26	0.42	-0.11	1.00		
Sr	0.77	0.02	0.37	0.17	0.23	0.10	-0.10	0.45	-0.19	0.19	1.00	
Ba	0.59	0.08	0.38	0.17	0.09	0.01	-0.17	0.32	-0.03	0.22	0.82	1.00

Table 5. Averages, standard deviations and ranges of conodont compositions

	Shale (5 grains, 15 analyses)		Chert (4 grains, 11 analyses)	
	Avr. & Stdev.	Range	Avr. & Stdev.	Range
SiO ₂ (wt%)	1.16 ± 1.82	0.02~5.66	0.29 ± 0.11	0.13~0.47
Al ₂ O ₃	0.45 ± 0.77	<0.01~2.53	0.05 ± 0.04	<0.01~0.15
MgO	0.16 ± 0.19	0.02~0.67	0.09 ± 0.08	0.02~0.25
MnO	0.09 ± 0.05	<0.01~0.17	0.06 ± 0.04	<0.01~0.14
CaO	52.93 ± 0.48	42.07~56.31	54.55 ± 0.62	53.61~55.35
P ₂ O ₅ /CaO	0.74 ± 0.02	0.71~0.77	0.74 ± 0.02	0.71~0.77

in cherts and those in shale. P₂O₅/CaO ratios range from 0.71 to 0.77 with an average of 0.74, which is close to ideal values of fluor-, chlor- and hydroxyl-apatite (Ca₅(PO₄)₃(F,Cl,OH))(P₂O₅/CaO = 0.759).

Barite X-ray image analyses show that the chert rarely contains minute grains (~20 µm) enriched in Ba and S (Fig. 5), whereas depleted in Si. They are most likely barite, although quantitative analyses were not successful.

SYNTHESIS AND CONCLUSIONS

Implications of accumulation of the bedded chert

From geochemical characteristics of raw and composite data and microanalyses described above, we infer that accumulation of Mn, Cu, Ba, Sr, and P in addition to Si in the bedded chert is closely related to non-lithogenic components. Non-lithogenic components in bedded cherts are assumed to be authigenic phases (e.g., oxides, sulphates and phosphates) and biogenic debris (e.g., opaline silica and phosphate represented by conodont). Formation of these phases is closely related to marine biogeochemical cycle that involves surface uptake, biomineralization, scavenging by siliceous tests at depths.

Although some other element such as Zn and K also show significant enrichment relative to lithogenic components, its causes have been still unclear, due to the lack of any remarkable relationships with other elements. In the following, we focus accumulation processes of Mn, Cu, Ba, Sr and P.

Manganese and Cu The composite data of chert-shale couplets indicate that the bedded cherts are substantially enriched in Mn and Cu. For both raw and composite data, MnO/TiO₂ and Cu/TiO₂ positively correlate with SiO₂/TiO₂ that semi-quantitatively reflects contribution of excess silica of biogenic origin (Figs. 3 and 4). These lines of evidence show that accumulation of Mn and Cu in the bedded chert was closely related to deposition of radiolarian tests.

It should be here cautioned that the close association of Mn and Cu accumulation with radiolarian tests does not always reflect directly biological uptake of these elements in the ocean surface, although these heavy metals are known to be fundamental nutrients and in fact their concentrations are significantly higher in some zooplankton species than in seawater (Masuzawa *et al.*, 1988; Piper, 1994). Manganese and Cu incorporated in organic tissues are mostly decomposed before settling to

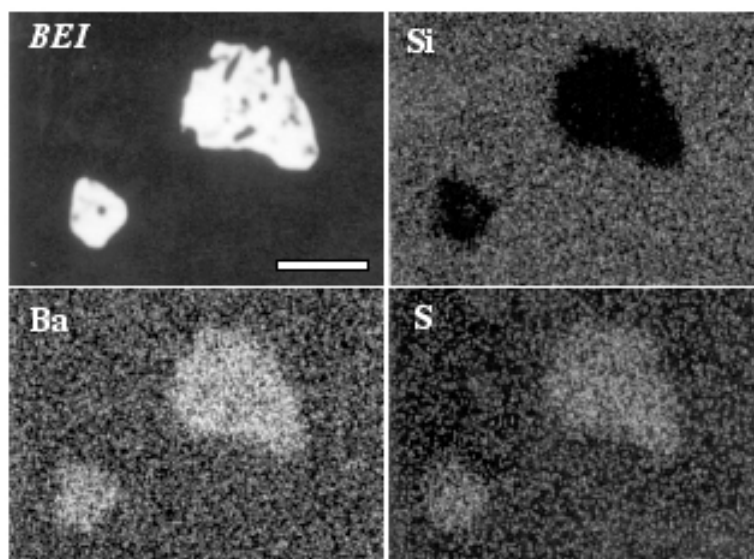


Fig. 5. X-ray images of Ba-rich particles. The scale the upper left is 10 μm . BEI: back-scattered electron image. Si: Image for Si-K α . Ba: Image for Ba-L α . S: Image for S-K α .

the ocean floor. Therefore, association of Mn and Cu with Si would require other mechanisms. It is implied that adsorption of Cu onto diatom frustules plays an important role for carrying Cu to the deep water (e.g., Boyle *et al.*, 1977; Sherrel and Boyle, 1992; Löscher, 1999). It is also suggested that some heavy metals including Mn and Cu can be incorporated into foraminiferal test or scavenged by particulate matter including biogenic particles (Boyle, 1981; Kuss and Kremling, 1999; Murray and Leinen, 1993; Martin and Knauer, 1985). Unfortunately, there have been little specific data on adsorption and incorporation of Mn and Cu to radiolarian tests in the modern ocean. Nevertheless, the following scenario could be made based on behaviors of Mn and Cu documented in the modern ocean system (Moffett, 1997; Dymond *et al.*, 1984; Baristrieri and Murray, 1986; Bishop and Fleisher, 1987). Mn^{2+} dissolved in the water column at least partially derived from organic tissues was oxidized to form suspended oxide particulate matter and/or scavenged onto radiolarian tests to form oxide-coatings, into both of which dissolved Cu^{2+} was incorporated or adsorbed. Furthermore, radiolarian tests, large fast-sinking particles, may have removed effectively these suspended small authigenic particles of metal-oxides (Sherrell and Boyle, 1992).

Barium and Sr Positive correlations of Ba/TiO₂ and Sr/TiO₂ with SiO₂/TiO₂ for both the raw and composite data set are very clear like Cu, indicating a close relationship of Ba- and Sr-accumulation with biogenic silica deposition. This inevitably suggests the presence of excess Ba and Sr in the bedded chert, which in particular for Ba, is supported by the presence of barite particles (Fig. 5),

major authigenic phase host for Ba in marine environment (Dehairs *et al.*, 1980; Bishop, 1988; Schmitz, 1987). As summarized by Bernstein and Byrne (2004), three hypothetical models for authigenic barite formation in oceanic water column are proposed. They are 1) formation within sulfate-enriched microenvironment such as diatom frustules (e.g., Bishop, 1988; Dehairs *et al.*, 1991), 2) formation mediated by dissolution of acantharian Ba-enriched celestite (SrSO₄) (Bernstein *et al.*, 1992) and 3) formation related to decomposition of phytoplankton, potential source for Ba (Ganeshram *et al.*, 2003). In the case of the bedded chert now concerned, it is implied from positive correlations of Sr with Si and Ba (Fig. 4, Table 4) that acantharian played an important role for barite formation.

Low enrichment factors of Ba and Sr (Table 3) are rather attributed to that their concentrations in hypothetical lithogenic components are much higher than those actually contained in the bedded cherts, in addition to some uncertainties in its original compositions. We consider that lithogenic materials in the bedded cherts were derived from significantly weathered and thus Ba- and Sr-depleted source. This is inferred from data of Ca, major alkali-earth element. In addition to the significantly low enrichment factor of Ca (E.F. < 0.3), Ca is contained mostly as apatite of diagenetic origin as discussed in the later section. Consequently, lithogenic components in the bedded cherts are depleted in Ca. This depletion is likely caused by chemical weathering of source rocks; during chemical weathering, Ba and Sr, as well as Ca were assumed to have been removed from source rocks.

Calcium and P Enrichment factor shows that the bedded

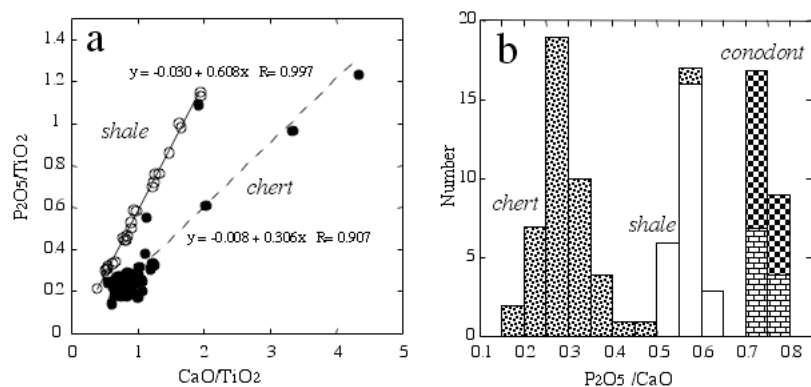


Fig. 6. a) CaO/TiO_2 - $\text{P}_2\text{O}_5/\text{TiO}_2$ plots for raw data. b) Histograms for $\text{P}_2\text{O}_5/\text{CaO}$ for cherts, shales and conodonts from cherts (brick) and shales (checker).

cherts are substantially enriched in P, relative to lithogenic components. Clear positive correlation between CaO/TiO_2 and $\text{P}_2\text{O}_5/\text{TiO}_2$ can be seen for cherts and shales (Fig. 6a); additionally the regression lines intercept nearly close to the origin. This shows that P and Ca in the bedded cherts are mostly contained in non-lithogenic fractions represented by apatite. Although conodont composed of apatite is ubiquitous in cherts and shales, the fossil is not a major host phase for P. Conodonts commonly have $\text{CaO}/\text{P}_2\text{O}_5$ ratios around 0.74, which are not identical to those of cherts (0.30 ± 0.07) and shales (0.57 ± 0.03) (Fig. 6b). Though not identified, their $\text{CaO}/\text{P}_2\text{O}_5$ ratios imply that the apatite species may be carbonate apatites ($\text{Ca}_5(\text{PO}_4)_3\text{-(CO}_3\text{,F)}_x(\text{F,OH,Cl})$). This apatite species would have $\text{CaO}/\text{P}_2\text{O}_5$ ratio of 0.51 and 0.25, if x takes 1 and 2, respectively. The values are relatively close to those of shales (0.57) and cherts (0.30); different $\text{CaO}/\text{P}_2\text{O}_5$ ratio between cherts and shales may be thus attributed to different carbonate number. Assumed presence of carbonate apatite is not unrealistic, because carbonate fluorapatite (CFA) is the dominant authigenic apatite formed during early diagenesis (e.g., Rutenberg and Berner, 1993; Van Cappellen and Berner, 1991), although the reason why apatites in cherts and those in shales have different carbonate content is unclear.

Implications of depositional environment and origin of bedded cherts

Depositional environment Geochemical features of Mn and Cu in the bedded cherts studied here can be explained by a relatively simple mechanism such as accumulation mediated by biogenic particles. This implies conservative behavior of these elements during early diagenesis, because if the elements were remobilized and regenerated significantly during early diagenesis, the primary geochemical signatures that reflect compositions of settling particles should have been lost. Behavior of Mn and

Cu during early diagenesis of marine sediments depends largely on the redox environment. These elements are in general less mobile under the oxic environment than the suboxic and anoxic one (Froelich *et al.*, 1979; Klinkhammer *et al.*, 1982; Sawlan and Murray, 1983). It is thus suggested that during the bedded chert formation, the depositional environment was oxic. Alternatively or additionally, the early-diagenetic elemental redistribution, which is exemplified by formation of surface oxidized Mn-rich zone (e.g., Pederesen *et al.*, 1986), was restricted within the distance corresponding to the thickness of proto-chert and proto-shale couplet, though less likely compared with the former interpretation.

Ba-Si relationship and implications for origin of bedded chert Excess-Ba contents, qualitatively expressed by Ba/TiO_2 in sediments, have been regarded as one of useful proxies for paleoproductivity (Dymond *et al.*, 1992; McManus *et al.*, 1998; Pfeifer *et al.*, 2001). This is based on that particulate Ba flux to sediments is controlled primarily by the surface production and that in general diagenetic regeneration is not significant (Dymond *et al.*, 1992; Paytan and Kastner, 1996). In analogy, accumulation of Ba in the bedded chert is possibly related to the fluctuation of surface productivity. If this is the case, the positive Ba-Si relationship indicates the accumulation of barite and biogenic silica without significant decoupling during sinking through water column and diagenesis.

Our interpretation for Ba-Si relationship appears to be inconsistent with recent knowledge about behaviors of biogenic particulate silica in modern ocean system (e.g., Tréguer *et al.*, 1995). In general, biogenic silica in sediments is not regarded as proxy for the surface productivity, because buried biogenic silica is assumed to be less than 5% of primary production due to active regeneration during settling and early diagenesis (e.g., Tréguer *et al.*, 1995), which is much smaller than Ba (~30%) (Paytan and Kastner, 1996). In the case of radiolarian

bedded cherts now concerned, on the other hand, blooming of radiolaria in the ocean surface may have directly resulted in increase in accumulation and burial of biogenic silica. This could be explained by assumed high productivity during the chert formation, like Tethyan radiolarites and by less dissolution susceptibility of radiolarian tests relative to diatoms dominant siliceous organisms in post-Cretaceous Ocean (De Wever and Baudin, 1996; De Wever *et al.*, 1995; Nelson *et al.*, 1996). De Wever *et al.* (1995) noted that in the zone of higher productivity, higher proportion of primarily produced silica could reach the sediments, and that high accumulation of biogenic silica could result in less dissolution at the sediment-water interface due to rapid burial. In addition to these conditions, the size of radiolaria should be important controlling factor. Like modern-ocean, Triassic seawater was likely under saturated with respect to opal. Therefore, if the travel time of particulate matter from the ocean surface to the bottom is shortened, regeneration of biogenic silica during sinking is reduced and significant portion of surface productivity would be buried. Radiolarian tests contained in the bedded cherts are up to 250 μm in diameter (fine-sand size), which is much larger than diatoms. It is believed that particulate opal composed of radiolarian tests had sunk much faster than modern equivalents originated from diatom, resulting significantly high ratio of buried-opal to surface production.

Hori *et al.* (1993) noted that the deposition of radiolarian bedded cherts in the Mino Belt reflects fluctuation of accumulation rate of siliceous tests (mostly radiolaria) with relatively constant clay sedimentation. Such simple models including an inverse case, i.e. fluctuation of accumulation rate of clay with relatively constant sedimentation of radiolarian tests, appear to be consistent with the results of present study, especially clear positive correlations between Si and Ba for both raw and integrated data. Other mechanisms interpreting bedded chert formation such as turbiditic re-deposition of siliceous sediments and significant silica redistribution during diagenesis (e.g., Nisbet and Price, 1974; Murray *et al.*, 1992b) are less likely in the present case. The present alternation of chert and shale parting is believed to reflect original cyclic sedimentation of siliceous and relatively clay-rich materials, though probably biased and exaggerated by later diagenetic redistribution of Si (De Wever *et al.*, 1995; Tada, 1991).

SUMMARY

The present study reveals sources of constituent elements in the Triassic radiolarian bedded cherts in the Mino Belt, central Japan. Examination of raw data, composite data of chert-shale couplets, and authigenic minerals suggest biogeochemical process played an important role for

the geochemistry of the bedded cherts, as summarized in the following.

- Significant portion ($\geq 50\%$) of Mn, Cu and P in the bedded cherts were supplied by non-lithogenic excess fractions.
- Positive correlation with $\text{SiO}_2/\text{TiO}_2$ suggests that accumulations of Ba, Sr, Mn and Cu were closely related to sedimentation of biogenic silica dominated by radiolarian tests.
- Manganese was probably accumulated as oxide particles and/or oxide coatings of radiolarian tests, into which Cu was incorporated. Barium and possibly Sr were accumulated mostly as authigenic barite. Host phase for excess-P is probably apatite, which involves three different species.
- Regeneration of redox-sensitive elements such as Mn and Cu is believed to have been not significant. Thus the depositional environment for the studied bedded chert section is suggested to be oxic.
- Alternation of chert and shale in the bedded cherts probably reflects original cyclic sedimentation of siliceous and relatively clay-rich materials.

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